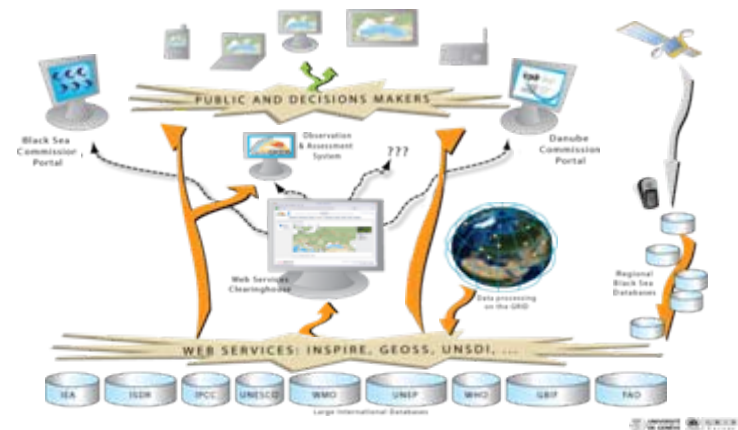
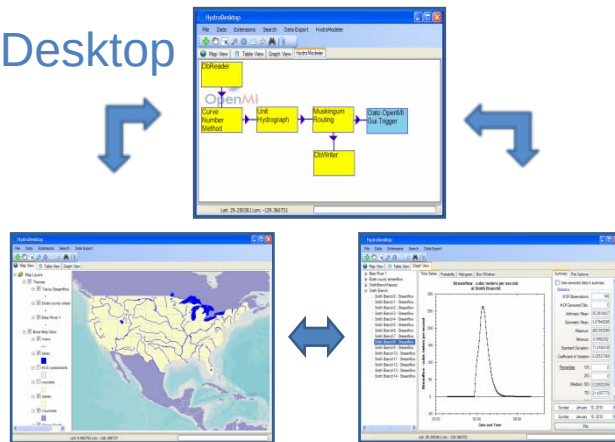


HydroDesktop



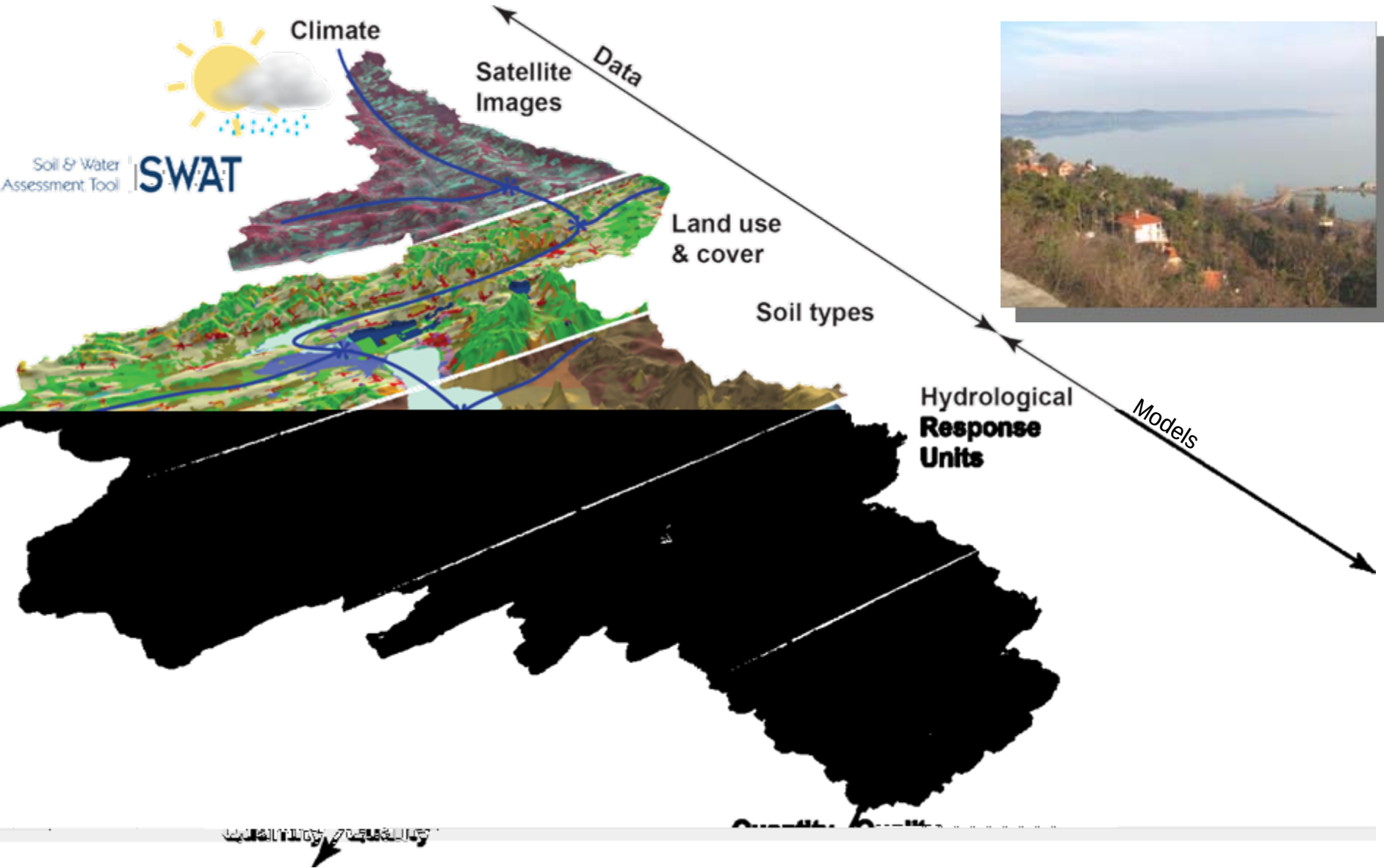
Potential Use of WaterML for hydrological modeling: SWAT, OpenMI, HYDRODESKTOP and EnviroGRIDS project

Ann van Griensven
UNESCO-IHE, Delft, Netherlands

Contributions from : David R. Maidment
Center for Research in Water Resources, University of Texas at Austin

Antony Lehmann
Université de Genève, Switzerland

Soil and Water Assessment Tool

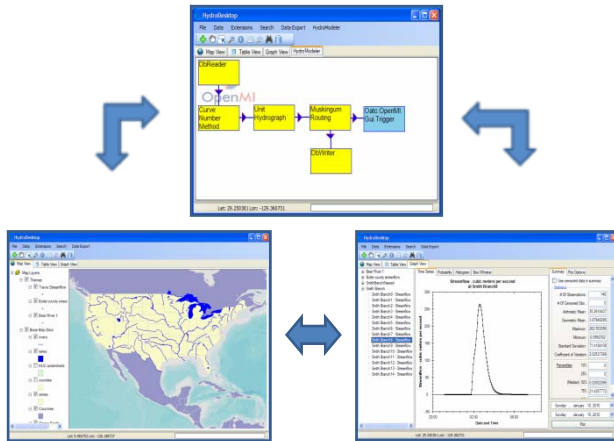


Swat-OpenMI

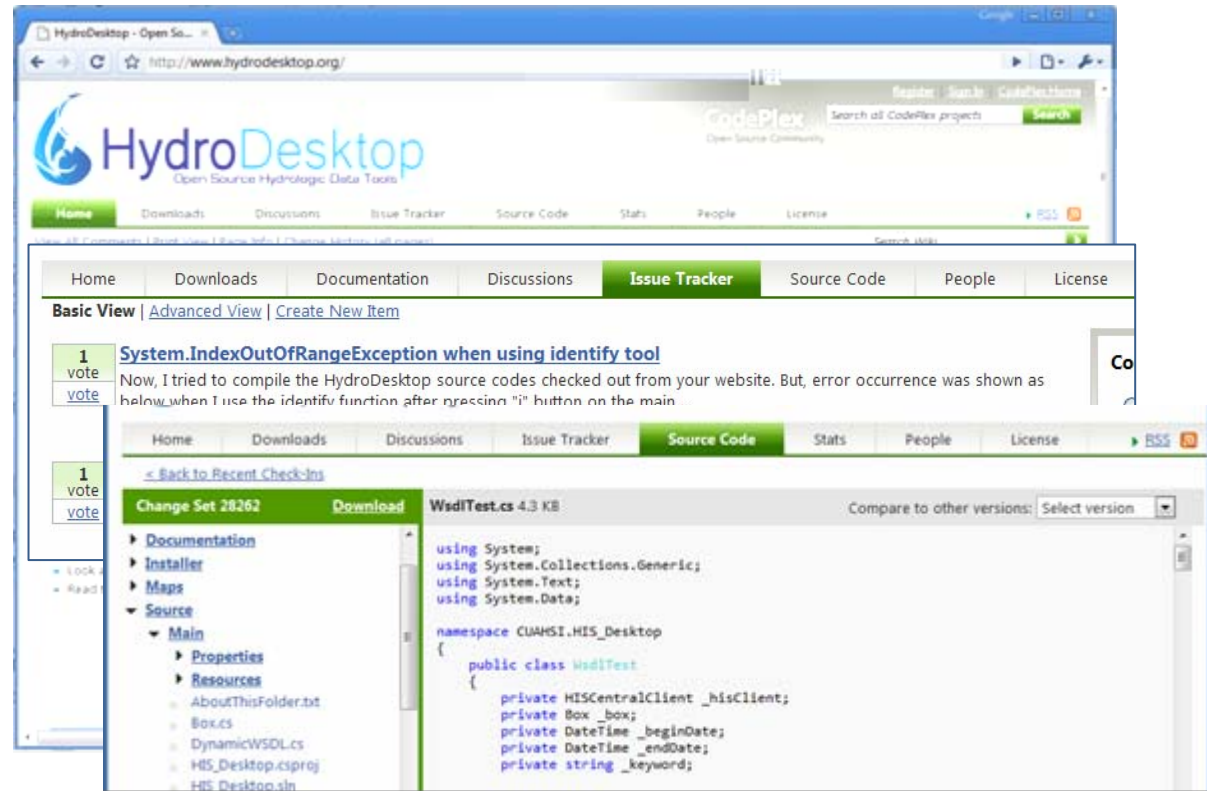
- Adaptations to the source code to make it an OpenMI component
- Link to:
 - Hydrodynamic model (sediment transport Blue Nile)
 - Link to climate models (US)
 - Integration into HydroDesktop (US)

<http://www.hydrodesktop.org>

A web-based collaborative design and development environment set up on CodePlex.com



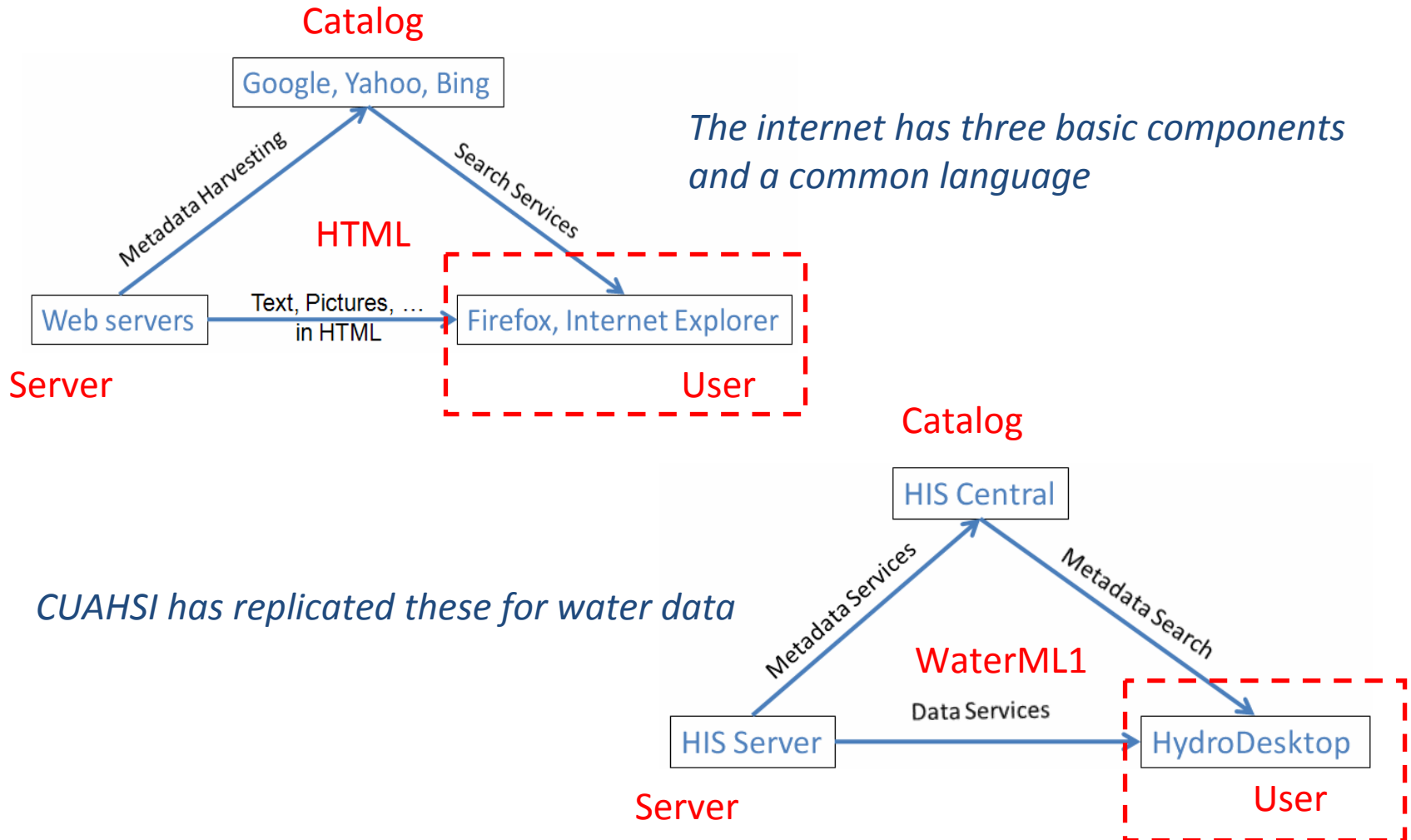
Alpha Version now
Beta in June 2010



Link to the OpenMI Association

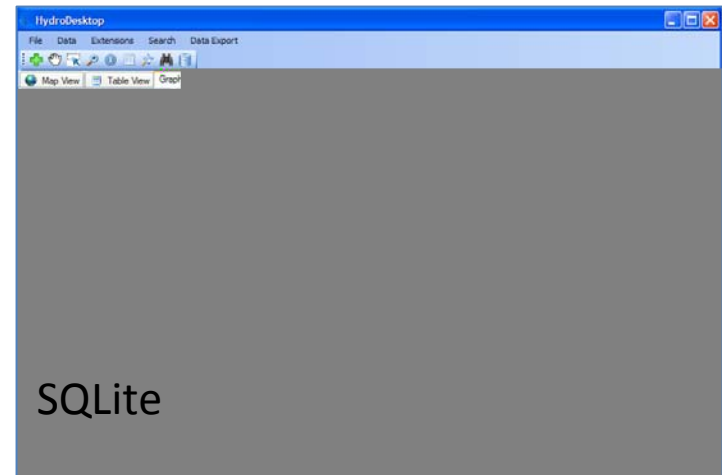
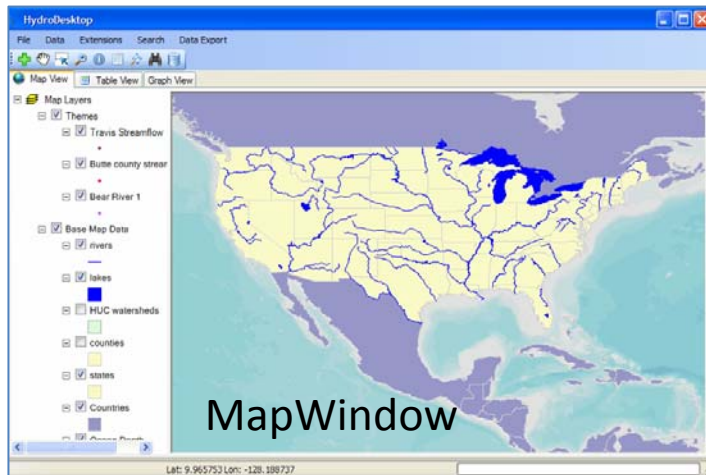
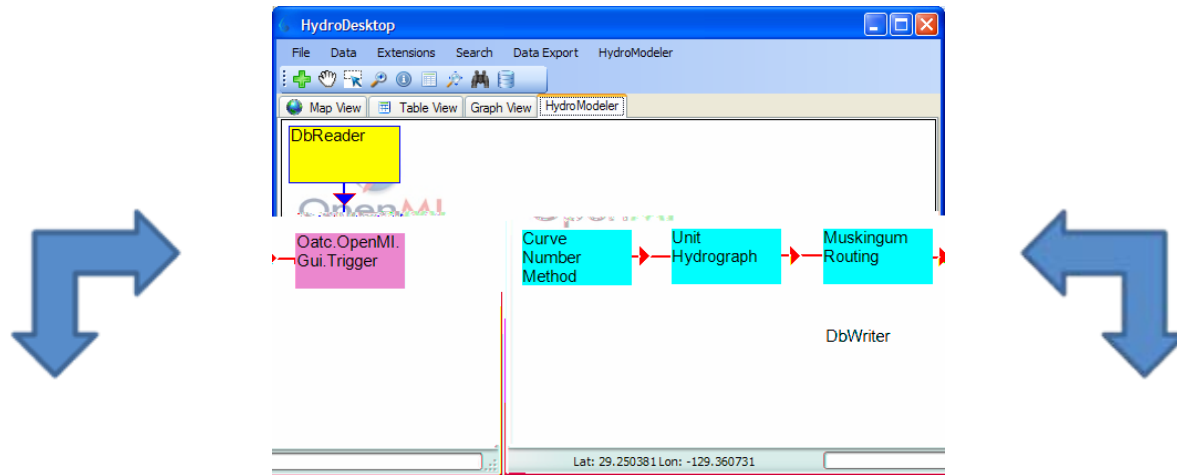
Water Web Services

A Hydrologic Information System integrates information from the internet



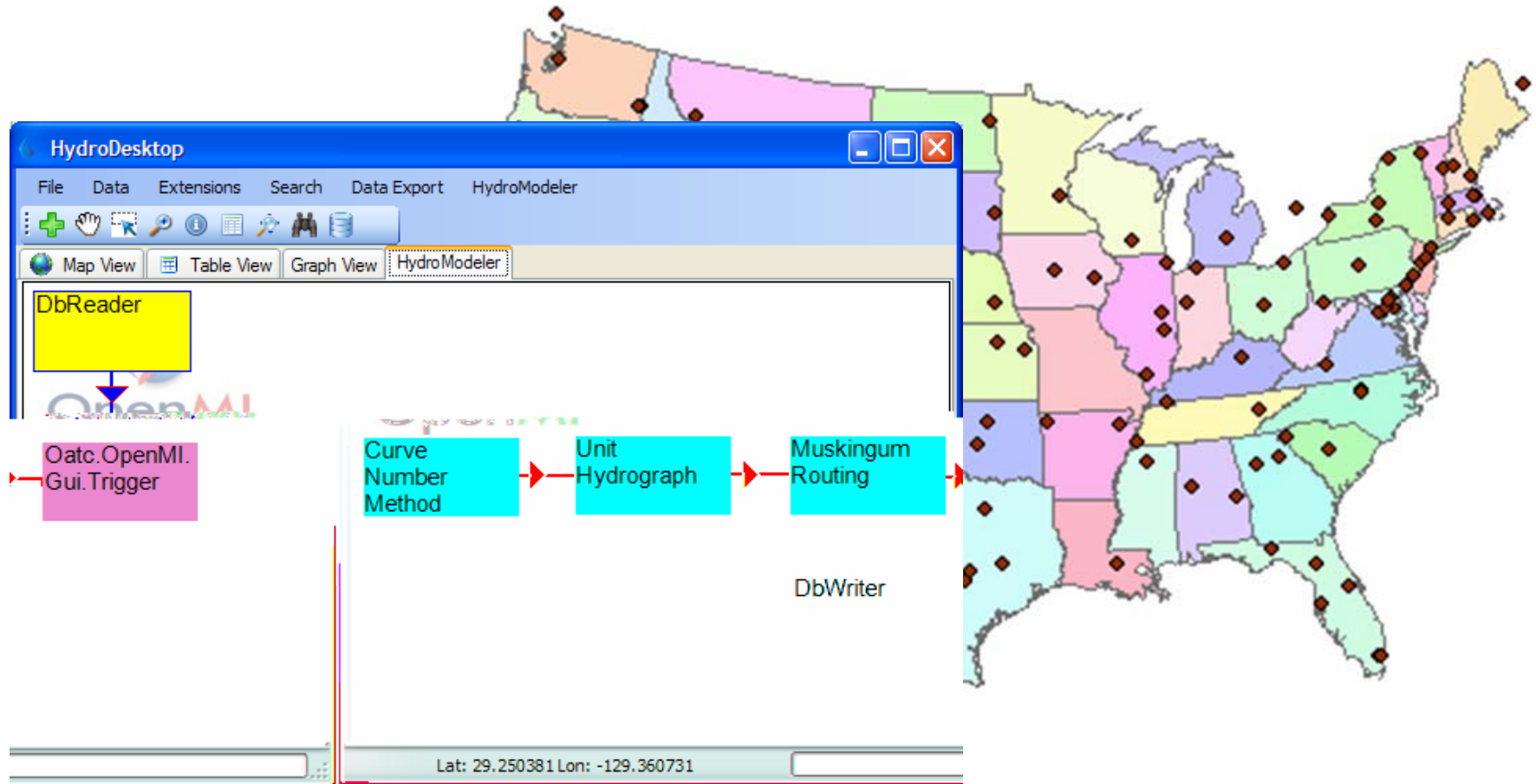
HydroDesktop

A Hydrologic Information System integrates space, time and modeling



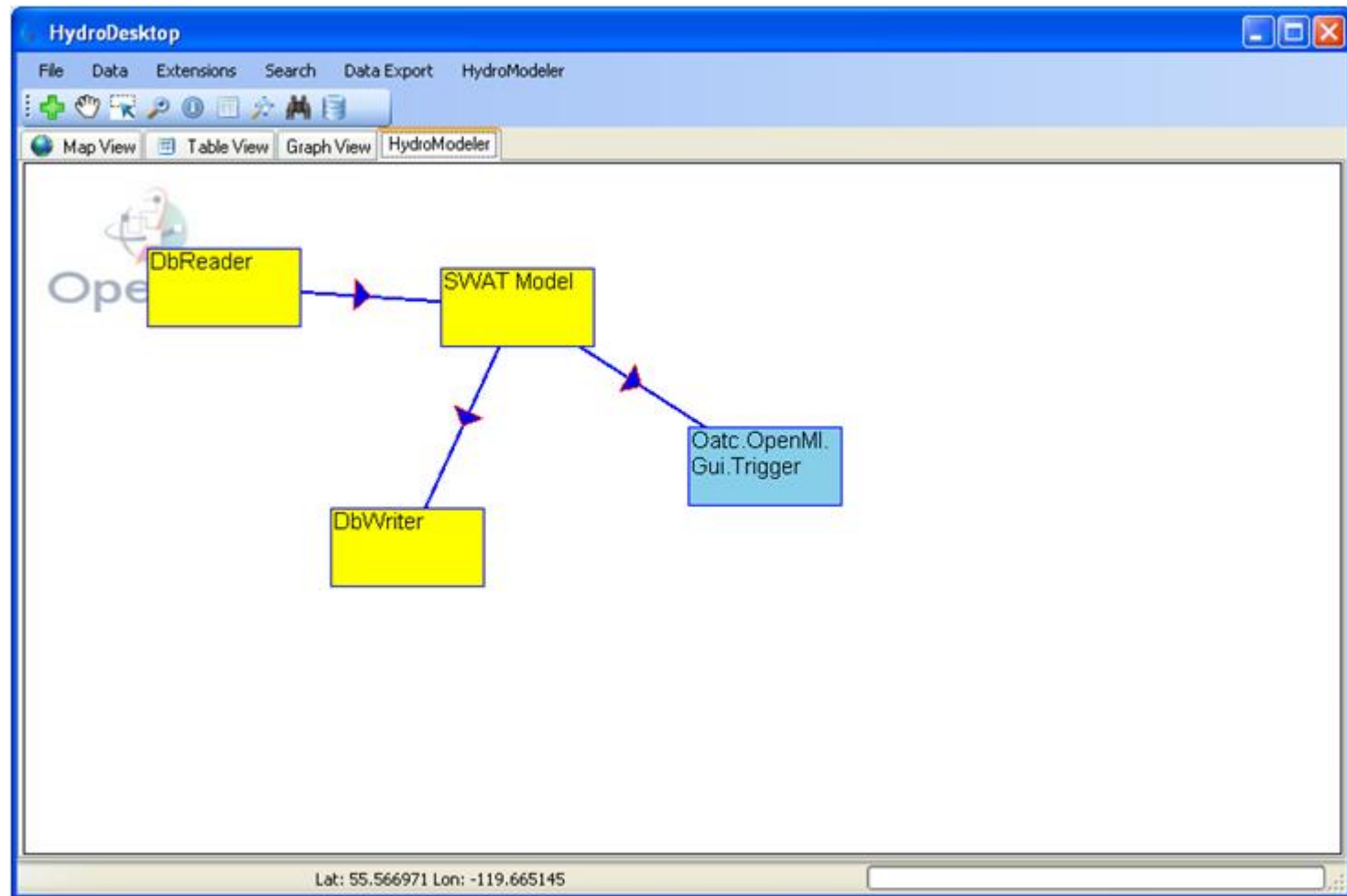
<http://www.hydrodesktop.org> – an open source software development

Library of OpenMI Components



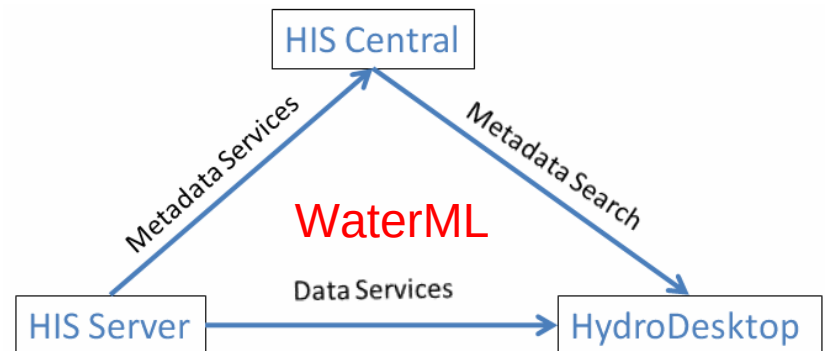
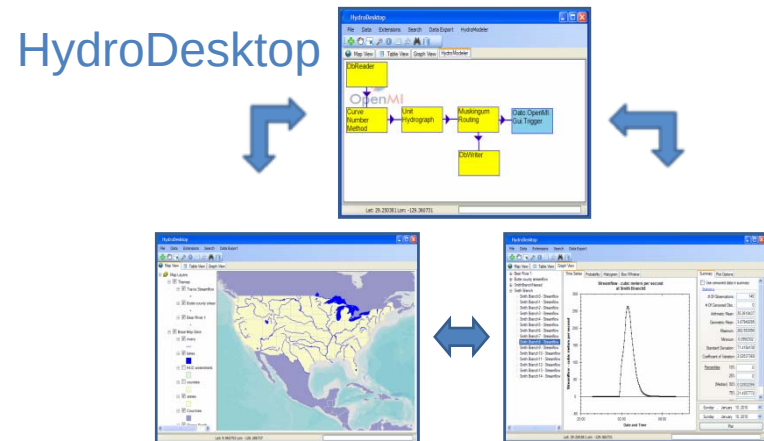
Invite the CUAHSI community to develop OpenMI components

Integrating SWAT



Conclusions

- Interoperability between **data and modeling** is critical
- A **hydrologic information system**
 - Integrates space, time and modeling
 - Integrates information from the internet
- **WaterML** is emerging as a standard for water time series data

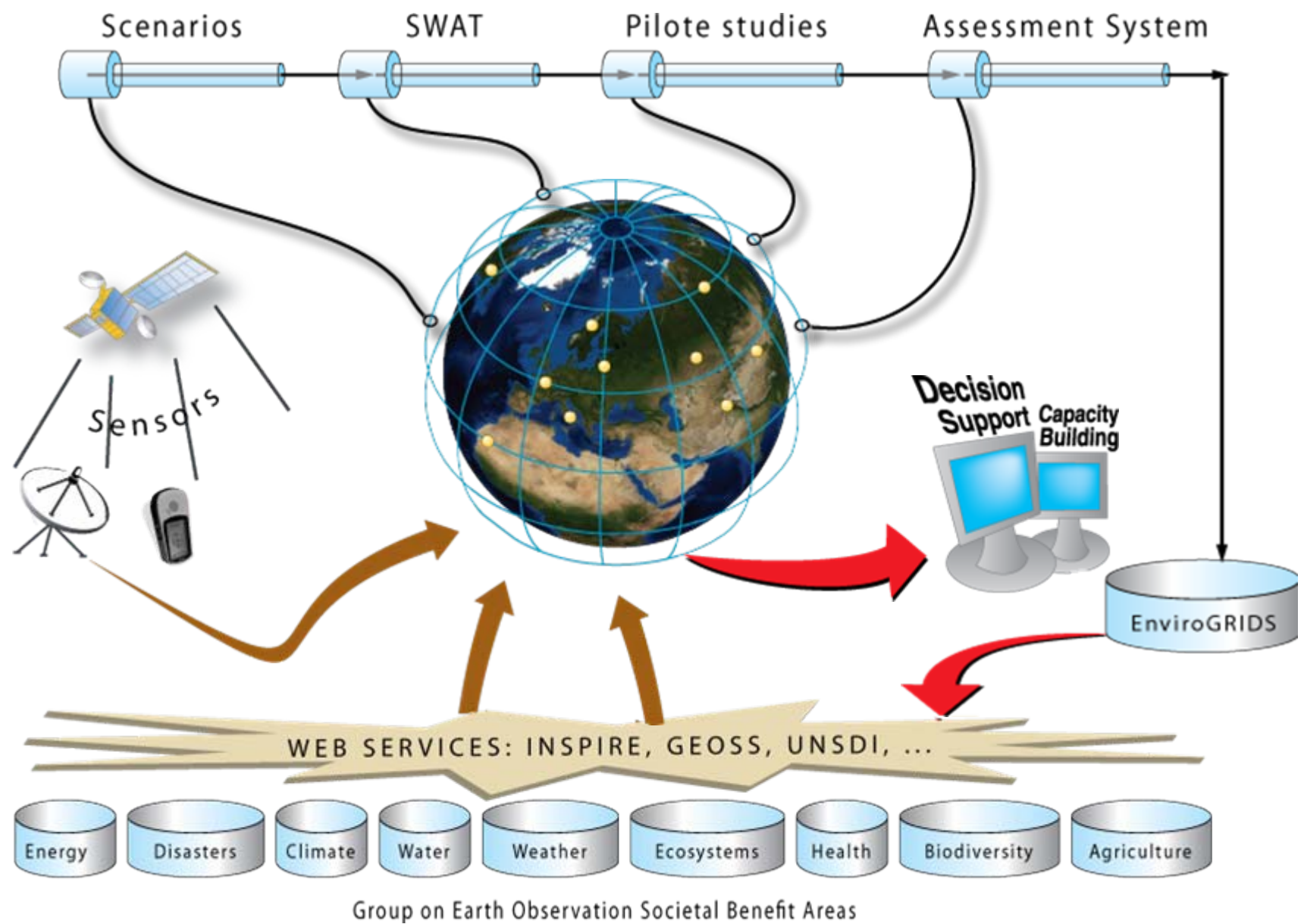


Building Capacity for a Black Sea Catchment Observation and Assessment System supporting Sustainable Development

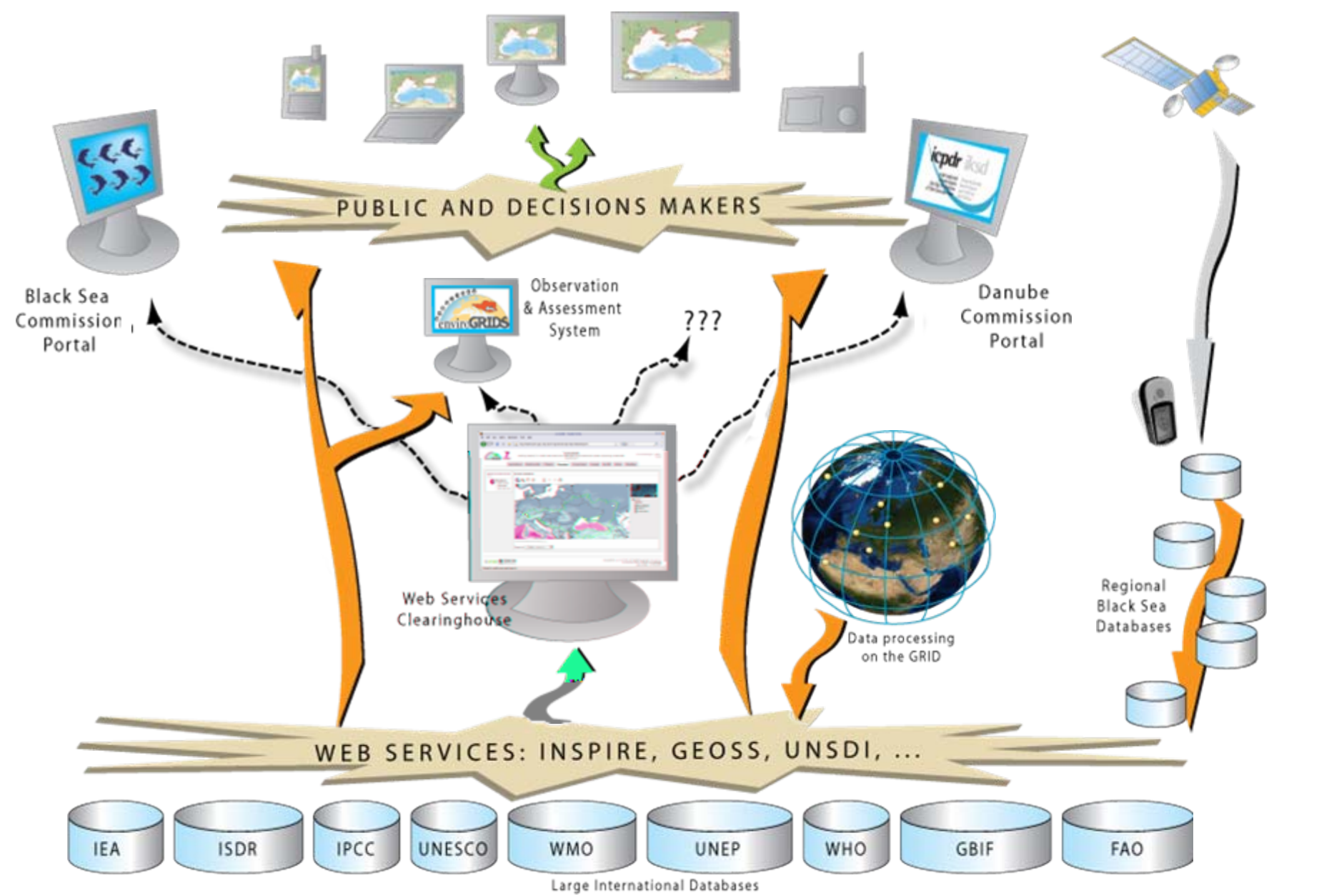
Dr Anthony Lehmann & enviroGRIDS consortium



enviroGRIDS: pipeline of analyses



enviroGRIDS: web services



Step 2: GRID technology

How can we store and process
large amount of data?



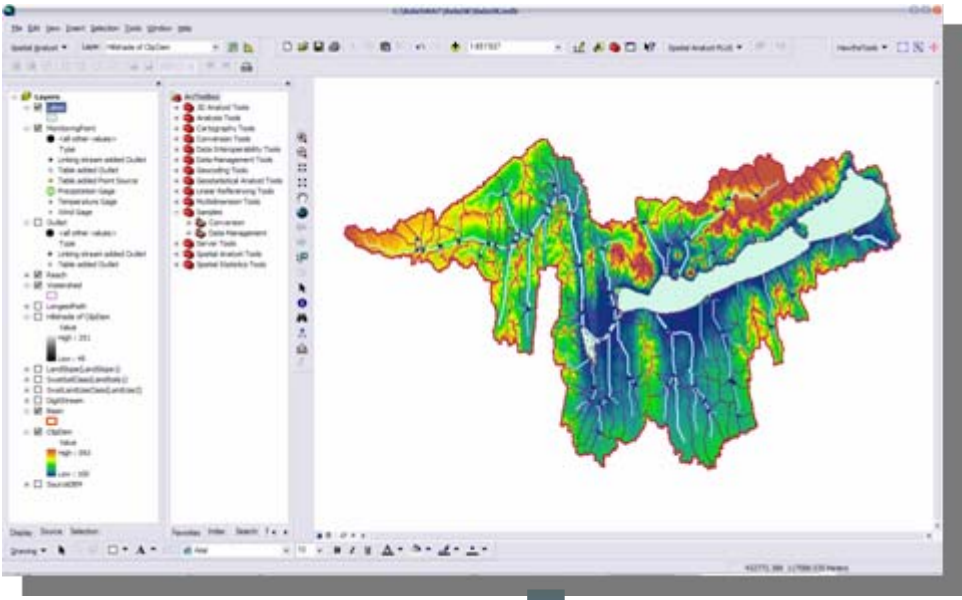




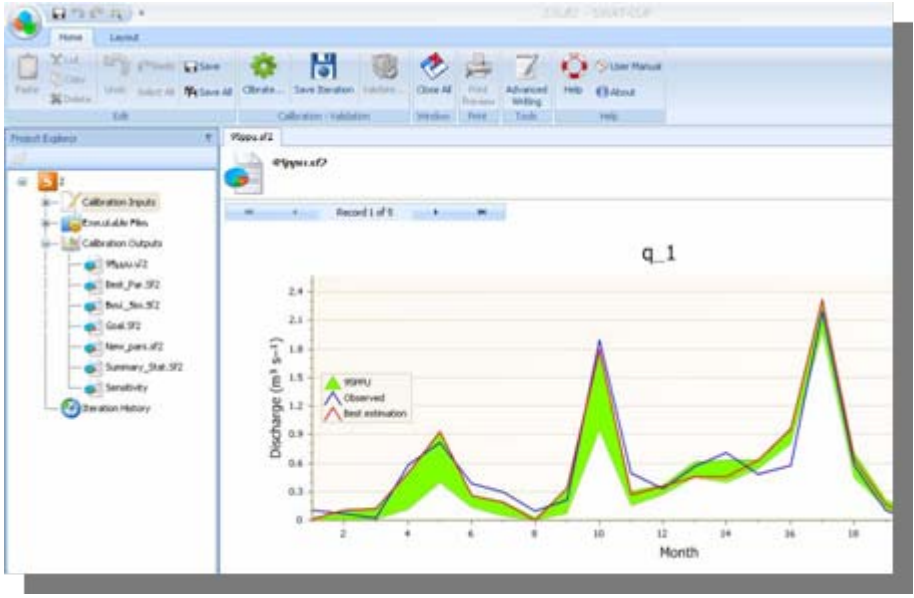
LHC Computing Grid project (LCG)



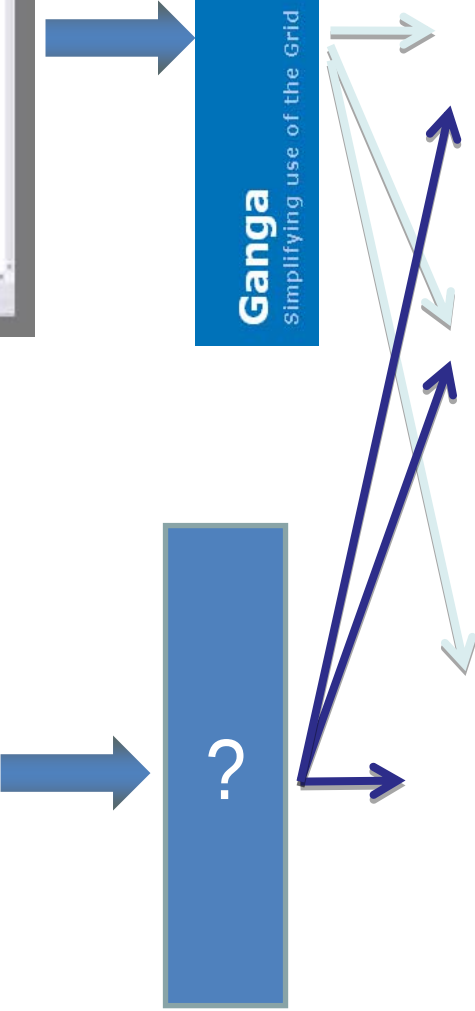
SWAT



SWAT-CUP



Ganga
Simplifying use of the Grid



How can we share
spatially explicit data?



SDI & GEOSS

 GROUP ON EARTH OBSERVATIONS

GEO Portal

provided by:
 

HOME | ABOUT GEOPORTAL | CONTACTS | NEWS | MAP VIEWER | | ADVANCED SEARCH | DOWNLOAD GEOPORTAL | ABOUT GEO

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 DISASTERS

 HEALTH

 ENERGY

 CLIMATE

 WATER

 WEATHER

 ECOSYSTEMS

 AGRICULTURE

 BIODIVERSITY



North America
Europe
Asia
Africa

zoom 

BREAKING NEWS

FLOODS IN IOWA, USA

Further to the flooding which occurred on 10 June, 10,000 people were forced to evacuate. The water is now moving towards Iowa City and Des Moines.



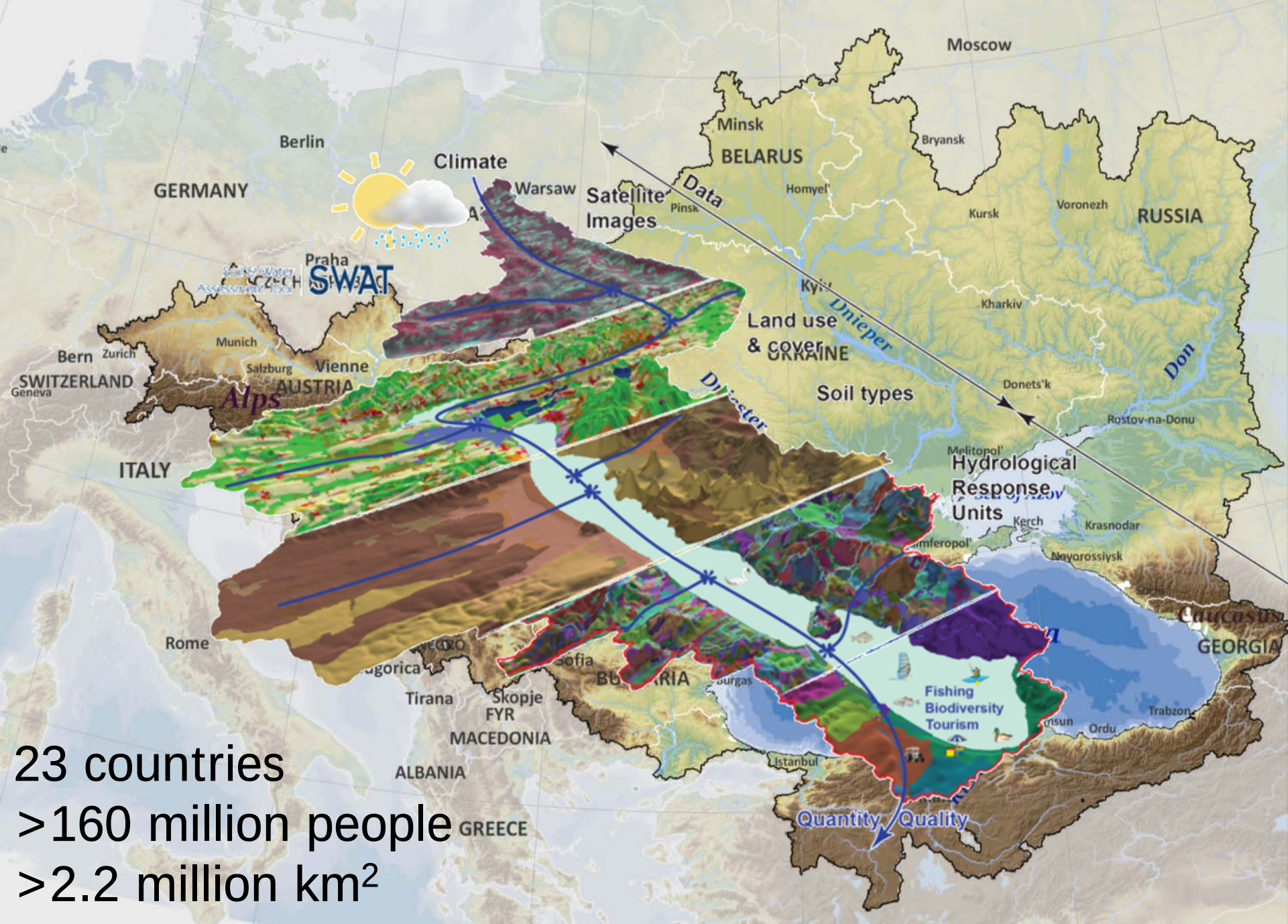
More...

WELCOME TO GEOPORTAL

The GEOportal provides an entry point to access remote sensing, geospatial static and in-situ data, information and services.

CAPACITY BUILDING 





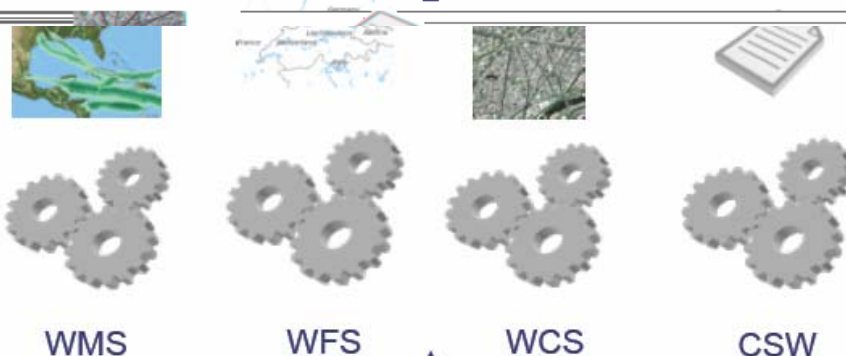


gSDI meeting – SDI model



Application layer

EG URM portal



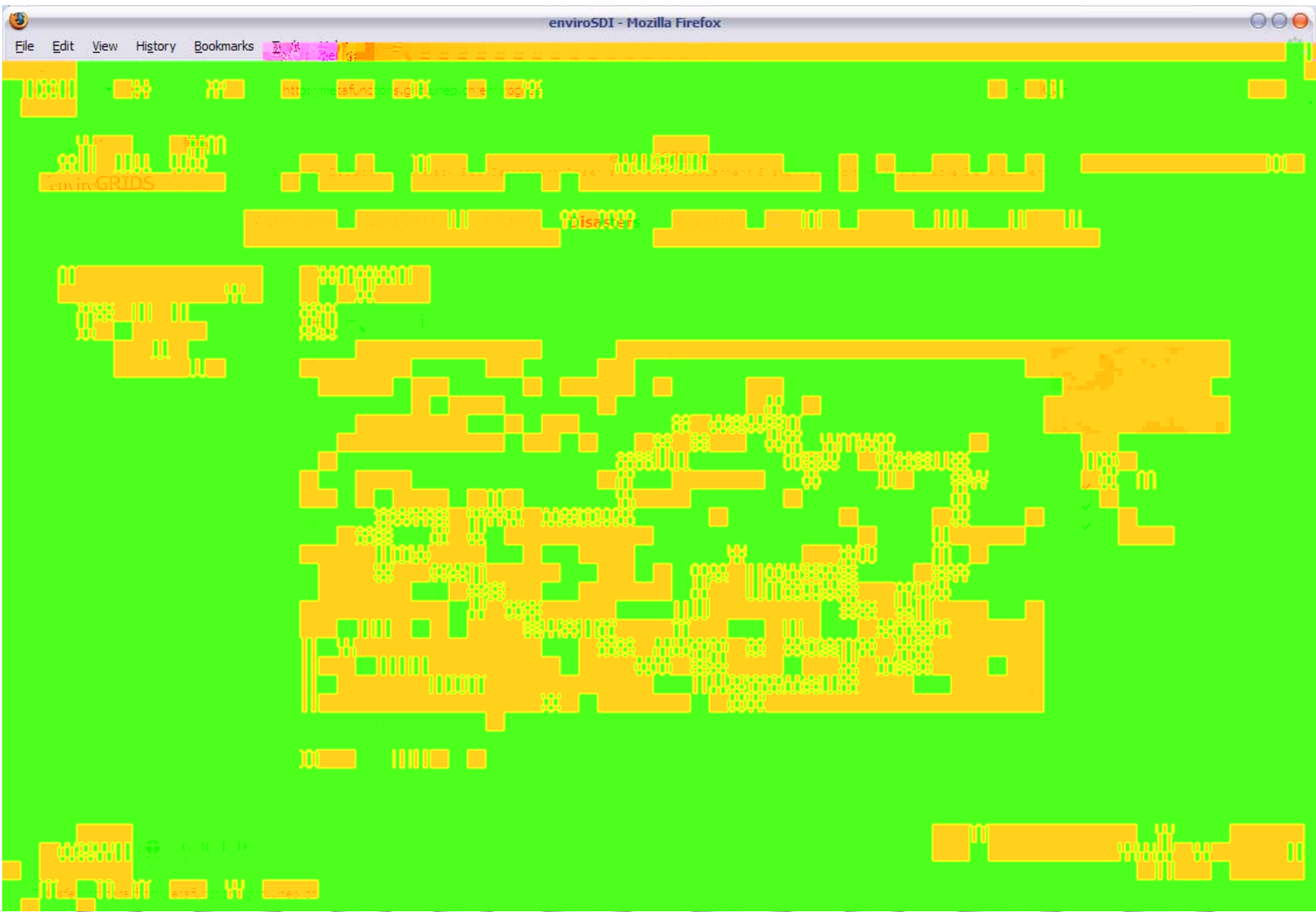
Service layer



Data layer

**Gap Analysis +
Tulcea Workshop**

Spatial Data & Services clearinghouse



Large International Databases

Challenges

- How to deal with time series data in the SDI?
- How to link to the modelling?
- How to link to the GRID?
- Many administrations, languages, ownership of data...